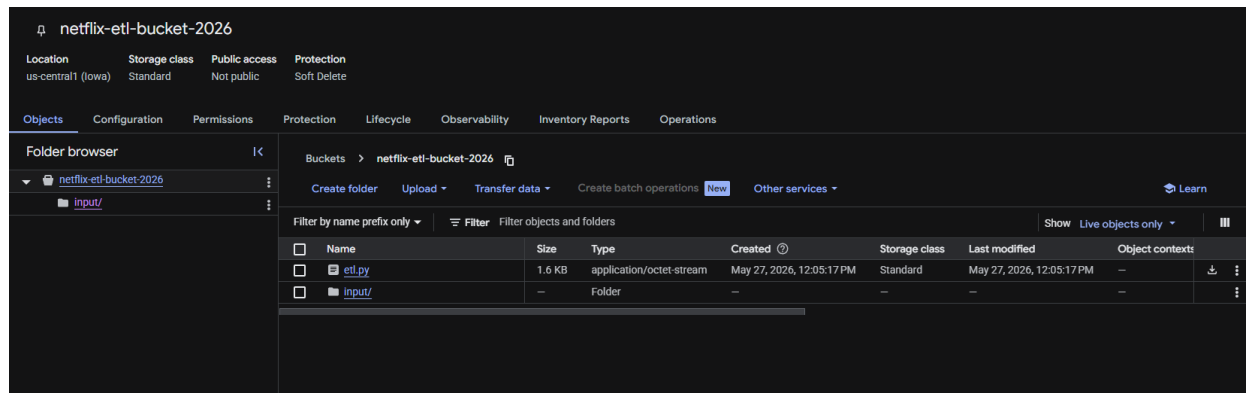


Google Cloud observation

1.1 The bucket was created and file was uploaded into input folder. Picture - 1



Picture -1

Here is pipeline code

```
schema = T.StructType([
    T.StructField("show_id", T.StringType(), True),
    T.StructField("type", T.StringType(), True),
    T.StructField("title", T.StringType(), True),
    T.StructField("director", T.StringType(), True),
    T.StructField("cast", T.StringType(), True),
    T.StructField("country", T.StringType(), True),
    T.StructField("date_added", T.StringType(), True),
    T.StructField("release_year", T.IntegerType(), True),
    T.StructField("rating", T.StringType(), True),
    T.StructField("duration", T.StringType(), True),
    T.StructField("listed_in", T.StringType(), True),
    T.StructField("description", T.StringType(), True)
])

df = spark.read.option("header", "true").schema(schema).csv("gs://netflix-etl-bucket-2026/input/")

df = df.filter(F.col("type") == "Movie")
df = df.filter(F.col("release_year") > 2016)
df = df.withColumn("duration_min", F.regexp_extract(F.col("duration"), r"(\d+)", 1).cast("int"))
df = df.filter(F.col("duration_min") > 90)

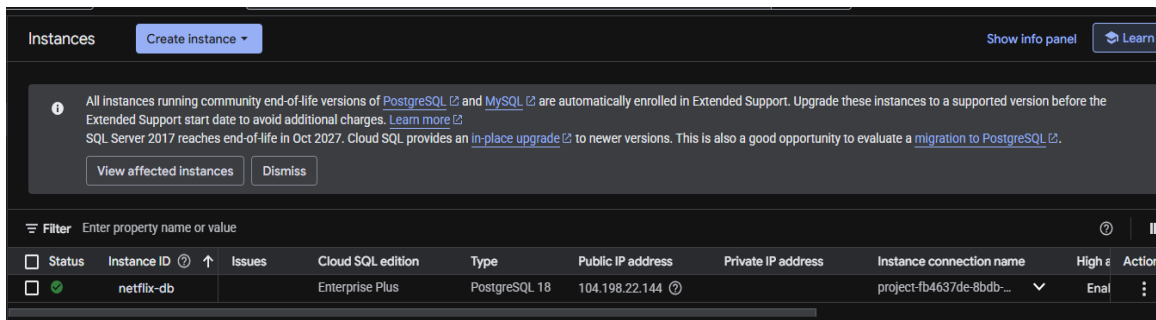
df_final = df.coalesce(1)

df.write \
    .format("jdbc") \
    .option("url", "jdbc:postgresql://104.198.22.144:5432/netflix_db") \
    .option("dbtable", "public.netflix_movies") \
    .option("user", "postgres") \
    .option("password", "password") \
    .option("driver", "org.postgresql.Driver") \
    .mode("overwrite") \
    .save()

from google.cloud import storage

client = storage.Client()
bucket = client.bucket("netflix-etl-bucket-2026")
blobs = bucket.list_blobs(prefix="input/")
for blob in blobs:
    blob.delete()
```

1.2 Instance was created (it's a kind of server so we could deploy our services)
Picture - 2



Picture - 2

1.3 I also created a cluster and then created a job(it's a kind of script) It's correct but it doesn't work since I there is some problem with billing. But I have tried to handle it many time and have checked everything but it still doesn't work. Perhaps it's a bug. As a result I decided to continue my exploring process by ignoring billing mistake. Picture (3,4)

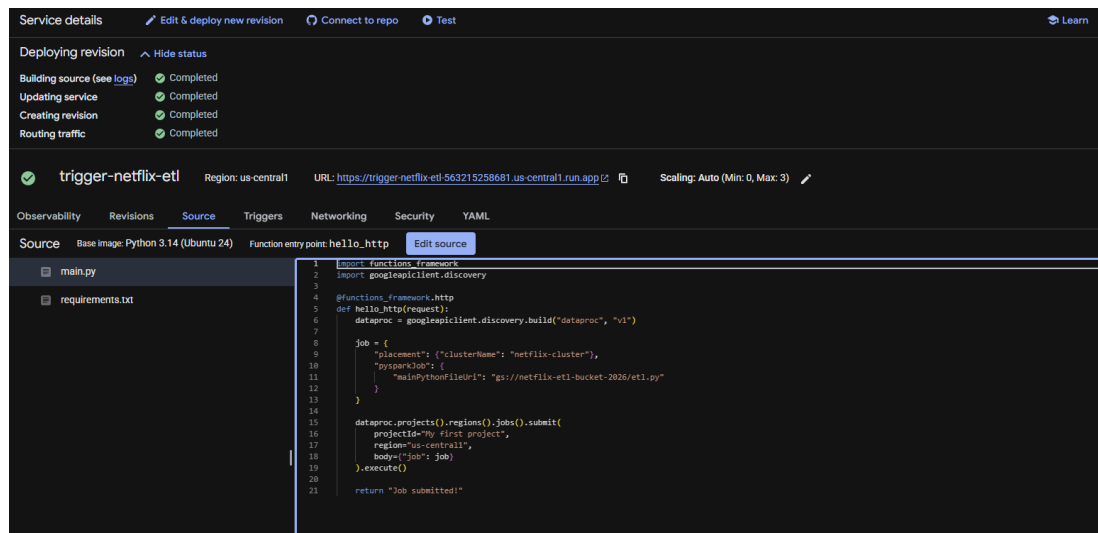
Name	Status	Region	Zone	Base image version	Engine	Total worker nodes	Flexible VMs?	Scale	Labels
netflix-cluster	Running	us-central1	us-central1-b	2.3.30-debian12	Default	2	No	Of	goog-datap... : enabled

Picture 3

Job ID	Status	Region	Type	Cluster	Start time	Elapsed time	Labels
job-340bed5e	Failed	us-central1	PySpark	netflix-cluster	May 26, 2026, 2:27:10 PM	8 sec	None

Picture 4

1.4 After that I built a Cloud run function (Pictures 5)



Picture – 5

1.4 Finally I set up a scheduler so my job could be performed every day at 12 pm.
Picture 6.

Cloud Scheduler / Jobs

Jobs Create job Refresh Force run Edit Copy Pause Resume Delete Learn

Scheduler jobs App Engine Cron jobs

Filter Filter jobs

☐	Name ↑	Status of last execution	Region	State	Description	Frequency	Target	Last run	Next run	Last updated	Actions
☐	netflix-etl-job	Has not run yet	us-central1	Enabled		0 0 * * * (America/Boa_Vista)	URL : https://trigger-netflix-etl-563215258681.us-central1.run.app/		May 28, 2026, 6:00:00 AM	May 27, 2026, 12:48:44 PM	⋮

As far as I understand it's extremely important so all services are created at the same region.